

Report

Atticus Tarleton

2026-06-05

The below code sets the project root to the right location, making file paths easier to manage

```
import os
from pathlib import Path

_project_root = Path.cwd()
while not (_project_root / ".git").exists():
    _project_root = _project_root.parent
    if _project_root == _project_root.parent:
        raise RuntimeError(".git not found - are you inside the repo?")
os.chdir(_project_root)
print(os.getcwd())
```

/Users/AGT/Desktop/ORCA/OkemoDataLearningProject

The below code block is going to create a database connection to the large and small clean healthcare databases made in the engineering.qmd file

```
import duckdb

con = duckdb.connect()
con.execute(f"""
    CREATE OR REPLACE TABLE small_healthcare AS
    SELECT * FROM read_parquet('{"Data/clean_data/small_2024_health_data.parquet"}')
""")
con.execute(f"""
    CREATE OR REPLACE TABLE large_healthcare AS
    SELECT * FROM read_parquet('{"Data/clean_data/large_2024_health_data.parquet"}')
""")
con.execute(f"""
```

```
CREATE OR REPLACE TABLE small_combined_healthcare AS
SELECT * FROM read_parquet('{"Data/clean_data/small_2024_health_data.parquet"}')
WHERE st_abbrev = 'VT' OR st_abbrev = 'US'
""")
```

<_duckdb.DuckDBPyConnection at 0x118d652f0>

Questions to answer 1) How does Vermont's ratios of health care professions look compared to the US as a whole - graph 1: Bar plot of ratio of vermont next to US ratio - will need to do some pivoting of data and such - graph 2: do the same as above, but instead by summing the northeast region - how does vermont compare to the local region - questions: - include vermont or not? - take the ratios of each state and average them, or sum the states values and average them? 2) States with the most physicians, and their total population - basically, do the numbers align - and do the same with the least physicians 3) States with the greatest ratio of physicians, and with the smallest ratio

This code chunk below sets up the datasets that will be plotted and used in the future

```
import pandas as pd

small_health_df = pd.read_parquet('Data/clean_data/small_2024_health_data.parquet')
large_health_df = pd.read_parquet('Data/clean_data/large_2024_health_data.parquet')

occupation_count_columns = small_health_df.columns[2:35].tolist()

small_ratio_df = small_health_df.copy()
for i in occupation_count_columns:
    small_ratio_df[i] = small_ratio_df[i]/small_ratio_df['popn_pums_23']

small_combined_ratio_df = small_ratio_df[(small_ratio_df['st_abbrev'] == 'VT') | (small_ratio_df['st_abbrev'] == 'US')]

small_combined_ratio_melted = small_combined_ratio_df.melt(id_vars='st_abbrev',
    value_vars=occupation_count_columns,
    var_name='occupation', value_name='ratio')
```

The code chunk below answers question 2 in pandas

```
physician_sorted_df = small_health_df.drop(51).sort_values(by='phys_wkforc_23', ascending=False)
print("\n the 10 states with the most physicians")
display(physician_sorted_df.head(10))
print("\n the 10 states with the fewest physicians")
display(physician_sorted_df.tail(10))
```

the 10 states with the most physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
4	CA	115435	39242785
32	NY	88345	19872319
43	TX	73675	29640343
9	FL	64392	21928881
38	PA	47128	12986518
35	OH	39774	11780046
13	IL	38946	12692653
21	MA	33466	6992395
22	MI	31279	10051595
33	NC	30755	10584340

the 10 states with the fewest physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
39	RI	4325	1095371
19	ME	4303	1377400
12	ID	3814	1893297
7	DE	3375	1005873
26	MT	3259	1105072
41	SD	2810	899194
45	VT	2354	645254
34	ND	2244	779361
1	AK	2158	733971
50	WY	960	579761

The code chunk below answers question 2 in SQL

```
print("\n the 10 states with the most physicians")
display(con.execute("""SELECT st_abbrev, phys_wkforc_23, popn_pums_23
    FROM small_healthcare WHERE st_abbrev != 'US'
    ORDER BY phys_wkforc_23 DESC LIMIT 10""").df())

print("\n the 10 states with the fewest physicians")
```

```
display(con.execute("""SELECT st_abbrev, phys_wkforc_23, popn_pums_23
FROM small_healthcare WHERE st_abbrev != 'US'
ORDER BY phys_wkforc_23 LIMIT 10""").df())
```

the 10 states with the most physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
0	CA	115435	39242785
1	NY	88345	19872319
2	TX	73675	29640343
3	FL	64392	21928881
4	PA	47128	12986518
5	OH	39774	11780046
6	IL	38946	12692653
7	MA	33466	6992395
8	MI	31279	10051595
9	NC	30755	10584340

the 10 states with the fewest physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
0	WY	960	579761
1	AK	2158	733971
2	ND	2244	779361
3	VT	2354	645254
4	SD	2810	899194
5	MT	3259	1105072
6	DE	3375	1005873
7	ID	3814	1893297
8	ME	4303	1377400
9	RI	4325	1095371

The code chunk below answers question 3 in pandas

```
small_ratio_df.sort_values(by='phys_wkforc_23', ascending=False)[['st_abbrev', 'phys_wkforc_23', 'popn_pums_23']]
```

	st_abbrev	phys_wkforc_23	popn_pums_23
8	DC	0.009578	672079
21	MA	0.004786	6992395
32	NY	0.004446	19872319
11	HI	0.003972	1445635
6	CT	0.003953	3598348
39	RI	0.003948	1095371
20	MD	0.003654	6170738
45	VT	0.003648	645254
38	PA	0.003629	12986518
37	OR	0.003590	4238714
35	OH	0.003376	11780046
7	DE	0.003355	1005873
5	CO	0.003228	5810774
30	NJ	0.003208	9267014
29	NH	0.003205	1387835
47	WA	0.003170	7740984
41	SD	0.003125	899194
19	ME	0.003124	1377400
22	MI	0.003112	10051595
13	IL	0.003068	12692653
51	US	0.003039	332387543
48	WV	0.003028	1784462
23	MN	0.002981	5713716
26	MT	0.002949	1105072
4	CA	0.002942	39242785
1	AK	0.002940	733971
9	FL	0.002936	21928881
42	TN	0.002913	6986082
46	VA	0.002912	8657499
33	NC	0.002906	10584340
34	ND	0.002879	779361
25	MO	0.002864	6168181
40	SC	0.002779	5212774
27	NE	0.002728	1965926
44	UT	0.002658	3331187
15	IA	0.002657	3195937
18	LA	0.002639	4621025
36	OK	0.002629	3995260

	st_abbrev	phys_wkforc_23	popn_pums_23
49	WI	0.002617	5892023
31	NM	0.002582	2114768
10	GA	0.002542	10822590
2	AZ	0.002511	7268175
43	TX	0.002486	29640343
0	AL	0.002462	5054253
14	IN	0.002348	6811752
17	KY	0.002261	4510725
16	KS	0.002207	2937569
3	AR	0.002141	3032651
12	ID	0.002014	1893297
24	MS	0.002010	2951438
28	NV	0.001910	3141000
50	WY	0.001656	579761

The code chunk below answers question 3 in SQL

```
con.execute("ALTER TABLE small_healthcare ADD phys_wkforc_ratio_23 FLOAT")
con.execute("UPDATE small_healthcare SET phys_wkforc_ratio_23 = phys_wkforc_23/popn_pums_23")
display(con.execute("SELECT st_abbrev, phys_wkforc_ratio_23, popn_pums_23 FROM small_healthcare"))
```

	st_abbrev	phys_wkforc_ratio_23	popn_pums_23
0	DC	0.009578	672079
1	MA	0.004786	6992395
2	NY	0.004446	19872319
3	HI	0.003972	1445635
4	CT	0.003953	3598348
5	RI	0.003948	1095371
6	MD	0.003654	6170738
7	VT	0.003648	645254
8	PA	0.003629	12986518
9	OR	0.003590	4238714
10	OH	0.003376	11780046
11	DE	0.003355	1005873
12	CO	0.003228	5810774
13	NJ	0.003208	9267014
14	NH	0.003205	1387835
15	WA	0.003170	7740984
16	SD	0.003125	899194

	st_abbrev	phys_wkforc_ratio_23	popn_pums_23
17	ME	0.003124	1377400
18	MI	0.003112	10051595
19	IL	0.003068	12692653
20	US	0.003039	332387543
21	WV	0.003028	1784462
22	MN	0.002981	5713716
23	MT	0.002949	1105072
24	CA	0.002942	39242785
25	AK	0.002940	733971
26	FL	0.002936	21928881
27	TN	0.002913	6986082
28	VA	0.002912	8657499
29	NC	0.002906	10584340
30	ND	0.002879	779361
31	MO	0.002864	6168181
32	SC	0.002779	5212774
33	NE	0.002728	1965926
34	UT	0.002658	3331187
35	IA	0.002657	3195937
36	LA	0.002639	4621025
37	OK	0.002629	3995260
38	WI	0.002617	5892023
39	NM	0.002582	2114768
40	GA	0.002542	10822590
41	AZ	0.002511	7268175
42	TX	0.002486	29640343
43	AL	0.002462	5054253
44	IN	0.002348	6811752
45	KY	0.002261	4510725
46	KS	0.002207	2937569
47	AR	0.002141	3032651
48	ID	0.002014	1893297
49	MS	0.002010	2951438
50	NV	0.001910	3141000
51	WY	0.001656	579761

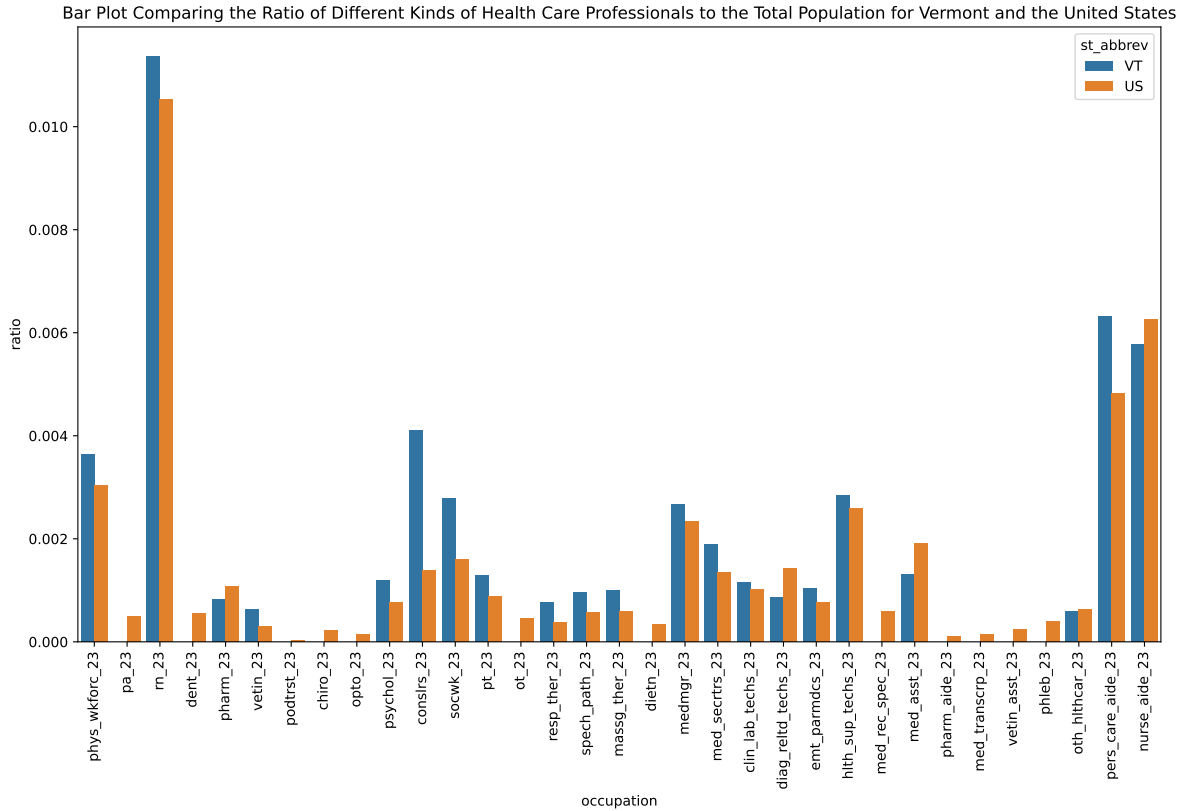
The code chunk below does the plotting for the first plot

What this plot is supposed to show is the ratio of different workers in different health care professions in Vermont compared to the US as a whole. This will be helpful in understanding how Vermont's healthcare system compares to the US.

For the results of this first plot, I am expecting to see that Vermont and the US have similar ratios of health care professionals to their population size. Some of this is because Vermont is part of the US, and so while there are substantive differences in culture across the US, I would assume that the rough amount of health care available is similar. Vermont also has the 19th highest median income of any state in the US ([See this article for more info](#)). I am assuming that median income roughly correlates to the number of health care professionals, as the higher the median income the more health care professionals a population can support in general. This is does have to do with the capitalist nature of private hospitals and systemic bias present in the US healthcare system, but it is an assumption I am making. Putting this together, because Vermont's median income is slightly higher than the average state, I am guess that Vermont has slightly ratio of health care professionals to its total population than the US as a whole, potentially with more professionals in fields that tend to have more elderly patients as well.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(14, 8))
plt.xticks(rotation=90)
sns.barplot(data=small_combined_ratio_melted, x='occupation', y='ratio', hue='st_abbrev')
plt.title("Bar Plot Comparing the Ratio of Different Kinds of Health Care Professionals to t")
plt.show()
```

Looking at the plot now, I do see what I expected to see. Vermont has more physicians than the US on average, and tends to have a slightly higher ratio of health professionals in general. While there are a number of professions where there is no blue bar, this is because there was no data collected for those professions, rather than there being basically no one practicing them. Besides the fact that the professions with no data in Vermont are generally the least numerous professions, I know that at least some of the lack of data is because it is not collected not that there is no one practicing those professions, as one bar with no data is `ot_23`, standing for occupational therapist and I know there are occupational therapists currently practicing in Vermont.